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Course/Section: CPE202/CPE31S1	**Date Submitted: 3/12/2026**
Instructor: Engr. Robin Valenzuela	**Semester and SY: Second Semester, S.Y. 2025-2026**

Midterm Skills Exam: Install, Configure, and Manage Log Monitoring tools

1. Objectives

Create and design a workflow that installs, configure and manage enterprise availability, performance and log monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

2. Instructions

1. Create a repository in your GitHub account and label it CPE_MIDEXAM_SURNAME.
2. Clone the repository and do the following:
   2.1. Create an Ansible playbook that does the following with an input of a config.yaml file and arranged Inventory file:
   2.2. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash) • Install Nagios in one host
   2.3. Install Grafana,Prometheus and Influxdb in seperate hosts (Influxdb,Grafana,Prometheus)
   2.4. Install Lamp Stack in separate hosts (Httpd + Php,Mariadb)
3. Document all your tasks using this document. Provide proofs of all the ansible playbooks codes and successful installations.
4. Document the push and commit from the local repository to GitHub.
5. Finally, paste also the link of your GitHub repository in the documentation.

3. Output (screenshots and explanations)

```yaml
---
- hosts: first_host
  become: True
  roles:
    - elastic_stack
    - nagios

- hosts: second_host
  become: True
  roles:
  - grafana_stack

- hosts: third_host
  become: True
  roles:
    - lamp_stack
```

- The image above is a screenshot of the finalized config.yml script

config.yml | main.yml elastic_stack/... | main.yml grafana_stack/... | main.yml la

elastic_stack > tasks > main.yml > { } 9 > { } ansible.builtin.template > mode

Ansible Tasks Schema - Ansible tasks file (ansible.json)

```yaml
#SPDX-License-Identifier: MIT-0
---
- name: check OS family
  ansible.builtin.set_fact:
    os_family: "{{ ansible_facts['os_family'] }}"

# ElasticSearch installation

- name: install dependencies for elasticsearch on ubuntu
  ansible.builtin.apt:
    name: "{{ item }}"
    state: present
    update_cache: True
  loop:
    - apt-transport-https
    - openjdk-11-jdk
    - curl
  when: os_family == "Debian"

- name: add elasticsearch GPG key on ubuntu
  ansible.builtin.apt_key:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: os_family == "Debian"

- name: add elasticsearch repo on Ubuntu
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

config.yml main.yml elastic_stack/... main.yml grafana_stack/... main.yml lamp_stack/...

elastic_stack > tasks > main.yml > { } 9 > { } ansible.builtin.template > mode

```yaml
- name: add elasticsearch GPG key on ubuntu

- name: add elasticsearch repo on Ubuntu
  ansible.builtin.apt_repository:
    repo: "deb https://artifacts.elastic.co/packages/{{ elasticsearch_version.spli
    state: present
  when: os_family == "Debian"

- name: install elasticsearch on ubuntu
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: True
  when: os_family == "Debian"

- name: install dependencies and repo for elasticsearch on centos
  ansible.builtin.yum:
    name:
      - java-11-openjdk
      - curl
    state: present
  when: os_family == "RedHat"

- name: add elasticsearch repo on centos
  ansible.builtin.yum_repository:
    name: elasticsearch
    description: "Elastic repository"
```

Inline Suggestions
All files
YAML

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

project2

config.yml | main.yml elastic_stack/... | main.yml grafana_stack/... | main.yml lamp_stack/...

elastic_stack > tasks > main.yml > { } 9 > { } ansible.builtin.template > mode

```yaml
- name: add elasticsearch repo on centos
  ansible.builtin.yum_repository:
    description: "Elastic repository"
    baseurl: "https://artifacts.elastic.co/packages/{{ elasticsearch_version.split
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true
  when: os_family == "RedHat"

- name: deploy elasticsearch config
  ansible.builtin.template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: root
    mode: '0644'

# Kibana installation

- name: install Kibana
  ansible.builtin.package:
    name: kibana
    state: present

- name: deploy Kibana config
  ansible.builtin.template:
```

```yaml
  - name: deploy Kibana config
    ansible.builtin.template:
      owner: root
      group: root
      mode: '0644'

# logstash installation

- name: install logstash
  ansible.builtin.package:
    name: logstash
    state: present

- name: deploy logstash config
  ansible.builtin.template:
    src: logstash.yml.j2
    dest: /etc/logstash/logstash.yml
    owner: root
    group: root
    mode: '0644'
```

- The images above are screenshots of the code for the elastic search stack.
- The longest part of the code belongs to elastic search. From taking the GPG key to adding the repo.
- Both Kibana and Logstash are included in the stack.

```
cluster.name: "ansible-cluster"
network.host: {{ elasticsearch_network_host }}
discovery.type: single-node
```

- The image above is a screenshot of the template file for elasticsearch

```
elastic_stack > templates > logstash.yml.j2
http.host: "0.0.0.0"
pipeline.workers: {{ logstash_pipeline_workers }}
```

- The image above is a screenshot of the template file for logstash.

```
elastic_stack > templates > kibana.yml.j2
server.port: {{ kibana_server_port }}
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://localhost:9200"]
```

- The image above shows a screenshot of the template file for kibana.

```
elastic_stack > vars > main.yml > logstash_pipeline_workers
#SPDX-License-Identifier: MIT-0
---
# vars file for elastic_stack
elasticsearch_version: "8.10.2"
kibana_version: "8.10.2"
logstash_version: "8.10.2"

elasticsearch_network_host: "0.0.0.0"
kibana_server_port: 5601
logstash_pipeline_workers: 2
```

- The image above is a screenshot of the variables file for the entire elastic search stack.

config.yml | main.yml .../vars | main.yml grafana_stack/... | main.yml lamp_stack/...

grafana_stack > tasks > main.yml > { } 14

Ansible Tasks Schema - Ansible tasks file (ansible.json)

```yaml
#SPDX-License-Identifier: MIT-0
---
- name: check OS family
  ansible.builtin.set_fact:
    os_family: "{{ ansible_facts['os_family'] }}"

#install dependencies
- name: install dependencies for Ubuntu
  apt:
    name: "{{ item }}"
    state: present
    update_cache: true
  loop:
    - wget
    - curl
    - gnupg
  when: os_family == "Debian"

- name: install dependencies for centos
  yum:
    name: "{{ item }}"
    state: present
  loop:
    - wget
    - curl
    - yum-utils
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

config.yml | main.yml .../vars | main.yml grafana_stack/... | main.yml lamp_stack/...

grafana_stack > tasks > main.yml > { } 14

```yaml
- name: install dependencies for centos
  loop:
    - yum-utils
  when: os_family == "Debian"

# influxdb
- name: add influxdata repo
  apt_repository:
    repo: 'deb https//repos.influxdata.com/{{ ansible_distribution | lower }} {{ a
    state: present
  when: os_family == "Debian"

- name: install influxDB on Ubuntu
  apt:
    name: influxdb={{ influxdb_version }}
    state: present
  when: os_family == "Debian"

- name: install influxdb on centos
  yum:
    name: influxdb-{{ influxdb_version }}
    state: present
  when: os_family == "Debian"

# prometheus installation
- name: download prometheus
  get_url:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

config.yml | main.yml .../vars | main.yml grafana_stack/... | main.yml lamp_stack/...

grafana_stack > tasks > main.yml > { } 14

```yaml
- name: install influxdb on centos
  when: os_family == "Debian"

# prometheus installation
- name: download prometheus
  get_url:
    url: "https://github.com/prometheus/prometheus/relaeses/download/v{{ prometheu
    dest: "/tmp/prometheus.tar.gz"

- name: extract prometheus
  unarchive:
    src: /tmp/prometheus.tar.gz
    dest: /opt/
    remote_src: true

- name: create prometheus user
  user:
    name: prometheus_version
    system: true
    shell: /sbin/nologin

- name: move prometheus binaries
  copy:
    src: "/opt/prometheus-{{ prometheus_version }}.linux-amd64/{{ item }}"
    dest: "/usr/local/bin/{{ item }}"
    mode: '0755'
  loop:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

grafana_stack > tasks > main.yml > { } 14

```yaml

- name: create prometheus directories
  file:
    path: "{{ item }}"
    state: directory
    owner: prometheus
    group: prometheus
  loop:
    - /etc/prometheus
    - /var/lib/prometheus

- name: move prometheus config and console files
  copy:
    src: "/opt/prometheus-{{ prometheus_version }}.linux-amd64/{{ item }}"
    dest: "/etc/prometheus/{{ item }}"
    owner: prometheus
    group: prometheus
  loop:
    - prometheus.yml
    - consoles
    - console_libraries

- name: create systemd device for prometheus
  copy:
    src: prometheus.service.j2
    dest: /etc/systemd/system/prometheus.service
    mode: '0644'
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

config.yml main.yml .../vars main.yml grafana_stack/... main.yml lamp_stac

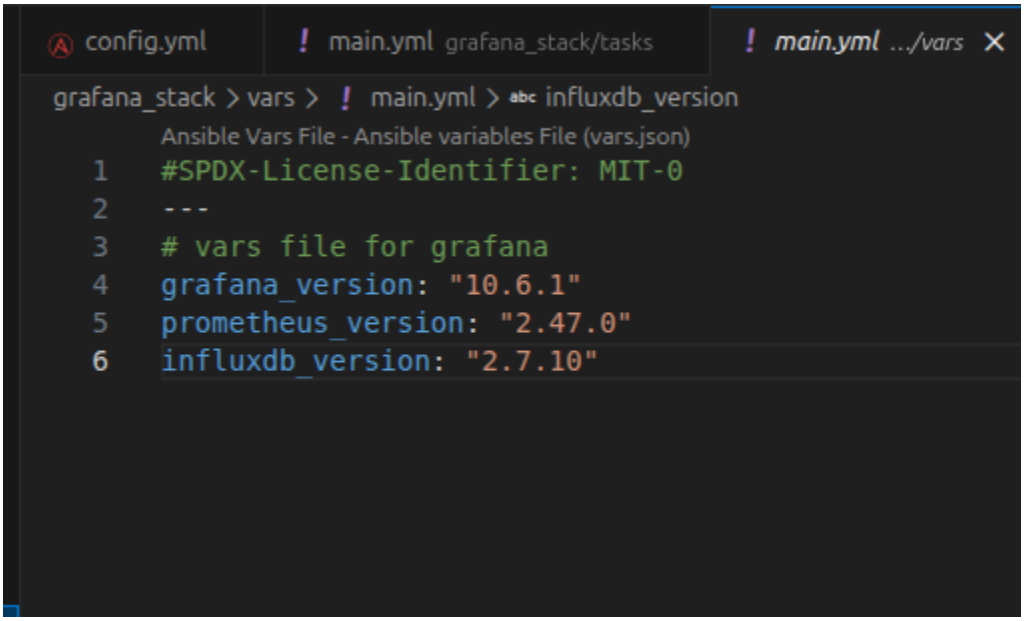
grafana_stack > tasks > main.yml > { } 14

```yaml
- name: create systemd device for prometheus


# grafana installation
- name: install grafana on ubuntu
  apt:
    name: grafana={{ grafana_version }}
    state: present
  when: os_family == "Debian"

- name: install grafana on centos
  yum:
    name: grafana-{{ grafana_version }}
    state: present
  when: os_family == "RedHat"
```

- The images above are screenshots of the code for the entire grafana, prometheus, and influxdb stack.
- The longest part of the code is prometheus, as it took a while to install as well.



- the image above is a screenshot of the variables file for the entire grafana stack.

yml | ! main.yml .../vars | ! main.yml grafana_stack/... | ! main.yml nagios/

lamp_stack > tasks > ! main.yml > { } 2

Ansible Tasks Schema - Ansible tasks file (ansible.json)

```yaml
#SPDX-License-Identifier: MIT-0
---
# tasks file for lamp_stack
- name: check OS family
  ansible.builtin.set_fact:
    os_family: "{{ ansible_facts['os_family'] }}"

# apache install
- name: install apache on ubuntu
  apt:
    name: "{{ apache_packages_ubuntu }}"
    state: present
    update_cache: true
  when: os_family == "Debian"

- name: install apache on centos
  yum:
    name: "{{ apache_packages_centos }}"
    state: present
  when: os_family == "RedHat"

```

lamp_stack > tasks > main.yml > { } 2

```yaml


# php installation
- name: install php on ubuntu
  apt:
    name: "{{ php_packages_ubuntu }}"
    state: present
  when: os_family == "Debian"

- name: install php on centos
  yum:
    name: "{{ php_packages_centos }}"
    state: present
  when: os_family == "RedHat"

# mariadb installation
- name: install mariadb on ubuntu
  apt:
    name: "{{ mariadb_packages_ubuntu }}"
    state: present
    when: ansible_os_family == "RedHat"

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE

g.yml | main.yml .../vars | main.yml grafana_stack/... | main.y

lamp_stack > vars > main.yml > [ ] mariadb_packages_ubuntu > abc 0

```yaml
# vars file for lamp_stack
apache_packages_ubuntu:
  - apache2
  - libapache2-mod-php

php_packages_ubuntu:
  - php
  - php-mysql

mariadb_packages_ubuntu:
  - mariadb-server
  - mariadb-client

apache_packages_centos:
  - httpd

php_packages_centos:
  - php
  - php-mysqlnd
```

- The images above are screenshots of the variables and main task file for the lamp stack.

```
g.yml    main.yml .../vars    main.yml grafana_stack/...    main.yml nagios/...
nagios > tasks > main.yml > { } 3
Ansible Tasks Schema - Ansible tasks file (ansible.json)
1  #SPDX-License-Identifier: MIT-0
2  ---
3  - name: detect OS family
4    ansible.builtin.set_fact:
5      os_family: "{{ ansible_facts['os_family'] }}"
6
7  # ubuntu installation
8
9  - name: install nagios and its dependencies on ubuntu
10   ansible.builtin.apt:
11     name: "{{ item }}"
12     state: present
13     update_cache: true
14   loop:
15     - nagios4
16     - nagios-plugins
17     - apache2
18     - php
19   when: os_family == "Debian"
20
21
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   ANSIBLE
serge@serge-VirtualBox:~/projects/project2$ ansible-playbook -i invent
```

- The image above is a screenshot of the tasks file for the nagios role.

```
single-node                 : ok=13    changed=0    unreachable=0    failed=0    skipped=4
   rescued=0    ignored=0
```

- screenshot of the succeeded elastic search stack.
- note: the other two stacks failed due to errors with the SSH keys during the setup of the other two virtual machines.

GitHub link:

https://github.com/Seijuro2004/MIDTERM.git

Conclusions: (link your conclusion from the objective)